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High-Performance Data Analytics Architecture for Meta Platforms

SECP3133-02 High Performance Data Processing

Week 4 Exercise

Topic:

Designing High-Performance Data Analytics Architectures

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1.0 Introduction

Meta Platforms is one of the world's largest social media tech giants, operating Facebook, Instagram, WhatsApp, and Threads, which are platforms used by over three billion people daily (Meta, 2024). Operating within the social media domain, Meta's core analytical focus centres on real-time user interaction and engagement analytics, including every like, share, comment, ad impression, and video view generated across its ecosystem. The sheer scale and speed of this data make high-performance data analytics not only advantageous but operationally essential. Meta must process hundreds of billions of events per day in near real time to power personalised news feeds, rank content, serve targeted advertisements, detect harmful content, and optimise platform engagement, all simultaneously and with sub-second latency (Meta Engineering, 2023). Without a high-performance data architecture integrating cloud computing, edge computing, and GPU-accelerated HPC, Meta would be unable to deliver the real-time personalised experiences that drive its business model and user retention.

Part A: Case Study Analysis

2.1 Data Characteristics

Meta Platforms operates one of the most complex and voluminous data environments in the world, spanning multiple data types, sources, and formats across its suite of social media products.

2.1.1 Types of Data

Meta handles all three major data types concurrently. Batch data is collected periodically for large-scale processing tasks such as daily user behaviour aggregation, ad performance reporting, and model retraining pipelines. Streaming data flows continuously from user activity feeds, representing an unending sequence of events that must be ingested and processed without interruption (Apache Kafka, 2024). Real-time data captures instantaneous interactions such as message delivery, live video reactions, and ad impressions, requiring immediate processing with sub-second latency to ensure a responsive user experience (Apache Flink, 2024).

2.1.2 Sources of Data

Data is produced across Meta's entire product portfolio. Facebook, Instagram, WhatsApp, Messenger, and Threads collectively generate billions of user interactions daily in the form of posts, comments, likes, shares, stories, and direct messages. Video uploads and live streams from Instagram Reels and Facebook Live contribute large volumes of unstructured media data. Advertisement events including impressions, clicks, and conversions are logged continuously across all platforms. System and application logs from user devices capture behavioural signals such as session duration, scroll depth, and notification responses, providing additional context for analytics and personalisation models (Meta Engineering, 2023).

2.1.3 The Three Dimensions of Data (3Vs)

Volume: Meta processes hundreds of billions of events daily across over three billion active users, producing petabytes of raw data that must be stored, indexed, and made queryable at scale (Meta Engineering, 2023).

Velocity: Data is generated every second of every day across all time zones and platforms simultaneously, requiring ingestion pipelines and processing frameworks capable of handling millions of events per second without bottlenecks (Apache Kafka, 2024).

Variety: Meta's data spans a wide spectrum of formats including text, images, videos, audio clips, system logs, social graph data, and structured advertising data, each requiring different storage strategies and processing techniques to extract meaningful analytical value (Meta Engineering, 2022).

2.2 Key Challenges

Meta's scale and operational complexity introduce several significant technical challenges that must be addressed through careful architectural design and technology selection.

2.2.1 Scalability

Meta must support over three billion active users across Facebook, Instagram, WhatsApp, Messenger, and Threads simultaneously. As the user base continues to grow, the underlying data infrastructure must scale horizontally to accommodate increasing volumes of events, media uploads, and social graph interactions without degradation in performance. A failure to

scale effectively would result in processing backlogs, delayed content delivery, and degraded user experience across all platforms (Meta Engineering, 2023).

2.2.2 Low Latency Processing

Meta's core products depend heavily on near-instantaneous data processing. News feed ranking, content recommendations, advertisement serving, and push notifications must all be computed and delivered within milliseconds of a triggering event (Apache Flink, 2024). Any processing delay directly impacts user engagement metrics and advertising revenue. This requirement necessitates a real-time processing architecture capable of evaluating and responding to millions of concurrent events per second across geographically distributed infrastructure.

2.2.3 Data Variety and Storage Management

Meta ingests data in a wide range of formats including text, images, videos, audio, system logs, social graph relationships, and structured advertising data (Meta Engineering, 2022). Each format carries different storage, indexing, and retrieval requirements. Managing this variety at petabyte scale introduces significant complexity in terms of data organisation, schema management, format compatibility, and cost-efficient tiered storage. A unified storage strategy must accommodate both structured and unstructured data while remaining accessible to downstream processing and analytics systems (Delta Lake, 2024).

2.2.4 AI Processing Cost and Resource Demand

Meta's recommendation engines, content ranking models, and content moderation systems are among the most computationally demanding AI workloads in the industry. Training and serving these large-scale deep learning models require substantial GPU and HPC resources, resulting in significant infrastructure cost and energy consumption. Efficiently scheduling training jobs, managing GPU cluster utilisation, and optimising model inference latency are ongoing operational challenges that require dedicated high-performance computing infrastructure (Meta AI Infrastructure, 2024).

2.3 Technologies Required

The following technologies form the core of Meta's high-performance data analytics infrastructure, each selected to address specific requirements identified in the data characteristics and challenges above.

2.3.1 Streaming Platform: Apache Kafka

Apache Kafka serves as Meta's primary real-time event streaming backbone. It ingests billions of events per second from across all platforms, acting as a distributed message queue that decouples data producers from downstream consumers (Apache Kafka, 2024). Kafka's high throughput, fault tolerance, and horizontal scalability make it well suited for handling Meta's continuous and unpredictable event load. It enables multiple downstream systems including stream processors, storage pipelines, and AI feature pipelines to consume the same event streams independently and concurrently (Apache Kafka, 2024).

2.3.2 Stream Processing: Apache Flink

Apache Flink is used for stateful, low-latency stream processing on data consumed from Kafka. It enables Meta to perform real-time computations such as feed ranking updates, anomaly detection, live engagement tracking, and fraud signal identification as events arrive (Apache Flink, 2024). Flink's ability to maintain processing state across time windows and deliver results within milliseconds makes it a critical component for all latency-sensitive analytical workloads.

2.3.3 Batch Processing: Apache Spark

Apache Spark handles large-scale batch processing workloads that do not require immediate results. This includes daily aggregation of user behaviour data, historical trend analysis, ad performance reporting, and periodic retraining of machine learning models (Apache Spark, 2024). Spark's distributed in-memory computation engine allows Meta to process petabytes of historical data efficiently, complementing Flink's real-time capabilities within a unified processing architecture.

2.3.4 Storage: Data Lake, Data Warehouse, and Data Lakehouse

Meta employs a multi-tier storage strategy to manage its diverse data assets. A data lake built on HDFS and object storage retains raw, unstructured, and semi-structured data at low cost for long-term retention. A Hive-based data warehouse organises structured and processed data for efficient analytical querying. Delta Lake functions as a data lakehouse layer, combining the scalability of a data lake with the ACID transaction guarantees and schema enforcement of a traditional warehouse, enabling both exploratory and production-grade analytics on the same dataset (Delta Lake, 2024).

2.3.5 AI and Machine Learning

Meta's AI infrastructure, built primarily around PyTorch, powers a broad range of models including personalised recommendation engines, content ranking models, natural language processing systems, computer vision pipelines, and content moderation classifiers (PyTorch, 2024). These models are trained on historical data at HPC scale and served in real time through optimised inference pipelines. Anomaly detection models also operate continuously to identify unusual patterns in user behaviour, system performance, and security events (Meta AI, 2024).

2.3.6 High Performance Computing: GPU Clusters

GPU clusters form the HPC backbone of Meta's AI operations. Large-scale model training workloads, including the training of LLaMA and recommendation models, require thousands of GPUs operating in parallel to complete within acceptable timeframes (Meta AI Infrastructure, 2024). Meta's HPC infrastructure enables distributed deep learning training using data parallelism and model parallelism techniques, significantly reducing training time compared to CPU-based alternatives and supporting the continuous model refresh cycles required for relevance and accuracy (Meta AI, 2024).

2.3.7 Cloud Infrastructure

Meta's cloud infrastructure provides the scalable foundation upon which all ingestion, processing, storage, and serving systems operate. Cloud-based auto-scaling ensures that compute and storage resources expand dynamically in response to traffic spikes, such as during global live events or viral content moments, without requiring manual intervention (Meta Engineering, 2023). Distributed cloud storage also ensures high availability and geographic redundancy across Meta's global data centres.

2.3.8 Edge Computing

Edge computing is employed at the client and network level to reduce round-trip latency for time-sensitive operations. User devices and nearby edge servers perform initial data capture, lightweight pre-processing, and local caching before forwarding events to Meta's centralised cloud infrastructure (Meta Engineering, 2022). This approach reduces the volume of raw data transmitted over the network, lowers perceived latency for end users, and ensures continued partial functionality even under degraded network conditions.

Part B: Architecture Design

3.1 Overview of Proposed Architecture

The proposed architecture for Meta Platforms is a hybrid high-performance data analytics system that integrates three computing paradigms, namely cloud computing, edge computing, and high-performance computing (HPC), into a unified and cohesive design. This hybrid approach was deliberately chosen to address the diverse and demanding requirements identified in the case study analysis, where no single computing paradigm is sufficient to handle the full spectrum of Meta's data processing needs.

Cloud computing forms the central backbone of the architecture, providing scalable and elastic infrastructure for data ingestion, stream and batch processing, storage, and output delivery. Edge computing is employed at the data collection and output layers, enabling low-latency event capture directly from user devices and nearby servers as well as fast content delivery to end users without unnecessary round trips to centralised data centres (Meta Engineering, 2022). High-performance computing, in the form of GPU clusters, powers the computationally intensive AI and machine learning workloads including model training, content ranking, and real-time inference that are fundamental to Meta's platform operations (Meta AI Infrastructure, 2024).

Together, these three paradigms complement one another across all six layers of the architecture, from data sources through to the output layer, ensuring that the system is capable of meeting Meta's requirements for scalability, low latency, data variety management, and AI processing efficiency as discussed in Section 2.2.

3.2 Architecture Diagram

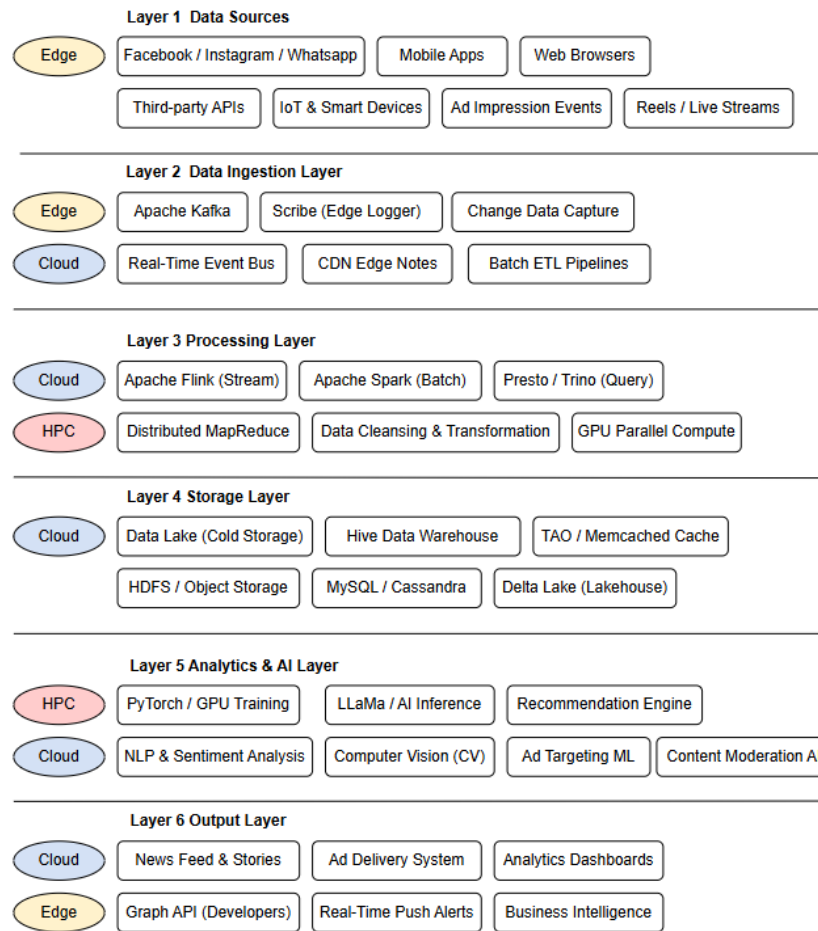


Figure 3.2 : High-Performance Data Analytics Architecture - Meta Platforms

3.3 Explanation of Architecture Layers

Layer 1 – Data Sources

Meta's data originates from billions of users across Facebook, Instagram, and WhatsApp, generating a continuous stream of interactions including likes, comments, shares, video views, and ad impressions (Meta, 2024). Additional sources include mobile apps, web browsers, third-party APIs, IoT-connected smart devices, and live streaming sessions via Reels and Facebook Live. This layer produces massive volumes of structured and unstructured data at extremely high velocity, representing the full variety dimension of big data.

Layer 2 – Data Ingestion Layer

Data is collected through two complementary pipelines. At the edge, Meta's proprietary Scribe system and CDN edge nodes capture and forward raw event logs from user devices with minimal latency (Meta Engineering, 2023). In the cloud, Apache Kafka serves as the central real-time event bus, ingesting billions of events per second (Apache Kafka, 2024). Change Data Capture (CDC) tracks database-level changes, while scheduled batch ETL pipelines handle large-scale historical data loads. This dual approach ensures both real-time responsiveness and completeness.

Layer 3 – Processing Layer

Ingested data is transformed through both stream and batch processing. Apache Flink handles real-time stream processing for time-sensitive tasks such as feed ranking and fraud detection, while Apache Spark manages large-scale batch processing for historical analytics (Apache Spark, 2024; Apache Flink, 2024). Presto/Trino enables interactive SQL queries across distributed datasets. For computationally intensive workloads, GPU clusters perform parallel processing at HPC scale, accelerating data transformation pipelines that feed downstream AI models (Meta AI Infrastructure, 2024).

Layer 4 – Storage Layer

Processed data is persisted across a multi-tier storage architecture. A Data Lake stores raw and semi-structured cold data cost-effectively, while a Hive Data Warehouse organises structured data for analytical queries. Delta Lake acts as a lakehouse, combining the flexibility of a data lake with the ACID transaction guarantees of a warehouse (Delta Lake, 2024). TAO and Memcached provide low-latency caching for Meta's social graph, and MySQL and Cassandra handle operational relational and wide-column data respectively (Meta Engineering, 2022).

Layer 5 – Analytics & AI Layer

This is the intelligence core of Meta's platform. PyTorch, developed by Meta, is used to train deep learning models on GPU clusters at HPC scale. LLaMA, Meta's large language model, powers inference tasks such as content understanding and generative features (Meta AI, 2024). A Recommendation Engine drives personalised feed and content ranking, while NLP and Sentiment Analysis models process textual content at scale. Computer Vision models enable

photo and video understanding, and Ad Targeting ML maximises ad relevance. A dedicated Content Moderation AI pipeline detects harmful content in near real time (Meta AI, 2024).

Layer 6 – Output Layer

Analytical results and model predictions are delivered to end users and downstream systems through multiple channels. The News Feed and Stories surface is dynamically ranked and served per user. The Ad Delivery System distributes targeted advertisements in real time. Processed insights are exposed via the Meta Graph API for third-party developers (Meta Engineering, 2023). Internal teams access metrics through Analytics Dashboards and Business Intelligence tools. At the edge, real-time push notifications are delivered directly to user devices with minimal latency, closing the loop from raw event to user-facing output.

Part C : Justification

4.1 Architecture Justification

The proposed hybrid architecture is well suited to Meta Platforms' operational requirements, as each design decision directly addresses the data characteristics and technical challenges identified in Sections 2.1 and 2.2.

The architecture supports real-time data processing through the combination of Apache Kafka in the ingestion layer and Apache Flink in the processing layer. Kafka continuously ingests billions of events per second from across all of Meta's platforms, while Flink processes these streams with sub-second latency to deliver timely outputs such as feed updates, ad impressions, and push notifications (Apache Kafka, 2024; Apache Flink, 2024). The inclusion of edge computing at both the data sources layer and the output layer further reduces end-to-end latency by capturing events close to the user and delivering results without unnecessary routing through centralised infrastructure.

The architecture is designed to scale horizontally across all layers. Cloud computing infrastructure in the ingestion, processing, and storage layers allows compute and storage resources to expand dynamically in response to demand. Distributed processing frameworks such as Apache Spark and Apache Flink are inherently scalable, capable of distributing workloads across large clusters to accommodate traffic growth without architectural changes

(Apache Spark, 2024). This ensures that the system remains performant as Meta's user base and data volumes continue to grow.

The processing layer explicitly separates streaming and batch workloads by assigning Apache Flink to handle real-time stream processing and Apache Spark to handle large-scale batch processing. This separation ensures that time-sensitive operations such as live feed ranking and anomaly detection are not disrupted by resource-intensive batch jobs such as historical aggregation and model retraining (Apache Flink, 2024; Apache Spark, 2024). The two pipelines operate concurrently within the same cloud infrastructure, allowing Meta to meet both immediate and long-term analytical requirements simultaneously.

The analytics and AI layer is purpose-built to support Meta's machine learning operations. PyTorch running on GPU clusters provides the HPC capacity required to train large-scale recommendation and content ranking models efficiently (PyTorch, 2024). Trained models are then deployed through optimised inference pipelines to serve personalised content, advertisement targeting, and content moderation decisions in real time. The inclusion of LLaMA for natural language understanding and computer vision models for media analysis further strengthens the architecture's AI capability across Meta's diverse content types (Meta AI, 2024).

The storage layer adopts a multi-tier strategy to accommodate Meta's wide variety of data formats. Raw and unstructured data such as video files, images, and system logs are retained in the data lake using HDFS and object storage, providing cost-effective long-term storage. Structured and processed data is organised within the Hive data warehouse for efficient analytical querying. Delta Lake serves as a lakehouse layer, bridging both tiers by enabling ACID-compliant transactions and schema enforcement on top of the data lake, ensuring that both exploratory and production analytical workloads can be served from a single unified storage platform (Delta Lake, 2024).

4.2 Computing Paradigm Justification

The selection of a hybrid architecture combining edge computing, cloud computing, and high-performance computing was driven by the specific operational demands of Meta's social media platform, where no single paradigm can address all requirements in isolation.

Edge computing was incorporated into the architecture primarily to minimise latency at the data collection and output stages. At the data sources and ingestion layers, edge nodes and Meta's Scribe logging system capture and pre-process raw event data directly from user devices and nearby servers before forwarding it to centralised cloud infrastructure (Meta Engineering, 2023). This reduces the volume of unprocessed data transmitted over the network and ensures that time-sensitive operations such as push notification delivery and content caching are handled as close to the user as possible. Edge computing is therefore essential for meeting the sub-second response times that Meta's platforms demand.

Cloud computing forms the central and most expansive tier of the architecture, spanning the ingestion, processing, storage, and output layers. Its primary advantage is elasticity, allowing Meta to scale compute and storage resources dynamically in response to unpredictable traffic patterns such as viral content events or global live broadcasts (Meta Engineering, 2023). Cloud infrastructure supports Apache Kafka for large-scale event ingestion, Apache Flink and Spark for stream and batch processing respectively, and a multi-tier storage strategy encompassing the data lake, data warehouse, and Delta Lake lakehouse (Apache Kafka, 2024; Apache Flink, 2024; Apache Spark, 2024). Additionally, cloud-hosted dashboards, business intelligence tools, and the Meta Graph API enable downstream consumers to access analytical outputs reliably and at scale. Cloud computing is therefore the most suitable paradigm for handling Meta's core data operations at a global level.

HPC in the form of GPU clusters was selected to address the computationally intensive demands of Meta's AI and machine learning operations. Training large-scale models such as LLaMA, deep recommendation engines, and computer vision classifiers requires parallel processing capabilities that neither edge nor cloud computing alone can provide cost-effectively (Meta AI Infrastructure, 2024). GPU clusters enable distributed deep learning training using data and model parallelism, significantly reducing training time and supporting the frequent model refresh cycles required to maintain recommendation accuracy and content relevance (Meta AI, 2024). HPC is therefore indispensable for sustaining the AI capabilities that are central to Meta's platform value.

The hybrid architecture was chosen because Meta simultaneously requires real-time responsiveness, global scalability, and powerful AI processing, three demands that map directly to edge, cloud, and HPC respectively. Relying on cloud computing alone would introduce unacceptable latency at the user-facing layer and insufficient GPU capacity for AI workloads. Relying solely on HPC would provide no elastic scaling for unpredictable traffic. Edge computing alone lacks the storage depth and processing power required for platform-wide analytics. By combining all three paradigms within a single cohesive architecture, the proposed design ensures that each layer operates at peak suitability for its specific function.

Part D: Presentation Summary

5.0 Presentation Summary

This section provides a concise overview of the key points that will be communicated during the group's oral presentation.

The selected case study is Meta Platforms, a global social media conglomerate operating Facebook, Instagram, WhatsApp, Messenger, and Threads (Meta, 2024). Meta's analytical focus is centred on real-time user interaction and engagement analytics, making it a highly relevant and technically demanding subject for high-performance data architecture design.

Meta's data environment is characterised by three major dimensions. In terms of volume, the platform processes hundreds of billions of events daily from over three billion active users, producing petabytes of data across all formats (Meta Engineering, 2023). In terms of velocity, data is generated continuously every second across all platforms and time zones simultaneously. In terms of variety, Meta handles text, images, videos, audio, system logs, social graph data, and structured advertising data, each requiring distinct processing and storage strategies.

The three major technical challenges identified are: scalability to support billions of concurrent users, low-latency processing for real-time feed ranking and ad delivery, and the high computational cost of training and serving large-scale AI models (Meta AI Infrastructure, 2024).

The proposed architecture is a six-layer hybrid system integrating edge computing, cloud computing, and HPC. Data flows from user devices and Meta's platforms through an edge-assisted ingestion layer, into cloud-based stream and batch processing pipelines, through a multi-tier storage system, and into an AI and analytics layer powered by GPU clusters, before results are delivered back to users and downstream systems through the output layer.

The architecture is suitable for Meta because it directly addresses each identified challenge. Edge computing ensures low-latency data collection and delivery, cloud computing provides the scalability and storage depth required for global operations, and HPC enables the AI model training and inference workloads that power Meta's core product experiences.

6.0 Conclusion

This report has presented a comprehensive high-performance data analytics architecture for Meta Platforms, designed to address the unique scale, speed, and intelligence requirements of one of the world's largest social media ecosystems. Through a structured case study analysis, the report identified the key data characteristics, technical challenges, and technology requirements that informed every layer of the proposed design.

The proposed hybrid architecture successfully integrates edge computing, cloud computing, and high-performance computing across six distinct layers, from data sources through to the output layer. By combining Apache Kafka and Flink for real-time ingestion and stream processing, Apache Spark for batch analytics, a Delta Lake-based multi-tier storage system, and PyTorch-powered GPU clusters for AI model training and inference, the architecture delivers a technically sound and practically feasible solution tailored specifically to Meta's operational environment (Apache Kafka, 2024; Apache Flink, 2024; Apache Spark, 2024; Delta Lake, 2024; PyTorch, 2024).

The proposed architecture enables Meta to achieve scalability across billions of concurrent users through elastic cloud infrastructure, low latency data capture and delivery through edge computing deployment, real-time analytics through dedicated stream processing pipelines, efficient storage of both structured and unstructured data through a unified lakehouse strategy, AI-based content recommendation and ranking through HPC-accelerated deep learning, and high-performance data processing across all layers of the system simultaneously.

In conclusion, the hybrid architecture proposed in this report demonstrates that no single computing paradigm is sufficient to meet the demands of a platform at Meta's scale. The deliberate combination of edge, cloud, and HPC paradigms, each assigned to the layers where it contributes most effectively, ensures that the architecture is both comprehensive in coverage and sound in its technical justification. This design serves as a practical reference for how

modern organisations can leverage high-performance data analytics to process massive volumes of data efficiently, accurately, and at speed.

7.0 References

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